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A Note on Relevance

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Abstract
In this paper, I present novel data that challenge the dominant approach to modelling linguistic relevance, according to which a sentence is relevant if and only if it is informative with respect to the Question Under Discussion (QUD). In particular, I show that a sentence can be uninformative with respect to the QUD and, despite this, still be perceived as relevant. In the final section, I outline the theoretical implications of this finding.

1 Introduction

In Gricean and Neo-Gricean approaches to pragmatics, utterances are assumed to be mediated by a series of conversational maxims. One such maxim is the Maxim of Relation (Grice's original terminology), also known as the Maxim of Relevance, which instructs speakers to make contributions that are related or relevant, in Grice (1975: 47) own words, 'to the immediate needs [...] of the transaction'. Grice left two issues open, however: (i) How should 'the immediate needs of the transaction' be characterised? (ii) What does it take for a linguistic contribution to count as relevant to 'the immediate needs of the transaction'?

Theoretical work carried out during the 80s and 90s—in particular, Groenendijk and Stokhof's (1984) ground-breaking work on questions and the seminal Roberts ([1996] 2012) —filled these two gaps in a rather compelling manner:

(i) Groenendijk and Stokhof (1984) identified '[t]he immediate needs [...] of the transaction' with 'the topic of the conversation', and the latter with (the meaning of) the explicit or implicit question that interlocutors are concerned with at a given point in discourse[1]—formally represented in their framework as a partition of a set of worlds. Since Roberts ([1996] 2012), this partition has been known as the *Question Under Discussion* (QUD for short) and the set of worlds over which it is defined has been identified with the context set, i.e. the set of worlds compatible with the common ground (Stalnaker 1978).

[1] Groenendijk and Stokhof (1984: 6): 'A notoriously difficult, but quite essential maxim proposed by Grice, is the Maxim of Relation. Relevance, it seems, is essentially tied to what a conversation is about, to what the topic of a conversation is. And a topic of conversation may very well be thought of as a (set of) question(s). This is obvious for discourses which consist of explicit question-answer sequences, but seems to hold also for types of conversation that are not explicitly concerned with information exchange. Even if in some discourse, no question is explicitly raised, it still plays an important role at the background, viz. as the topic that makes the discourse a coherent whole, rather than a random sequence of assertions. *The topic, i.e. an explicitly or implicitly raised question, is what defines the relevance of the assertions in a discourse for each other* [italics mine].'

(ii) Groenendijk and Stokhof (1984) conceptually linked the notion of relevance to that of question answering: a sentence is relevant with respect to a QUD Q if and only if it resolves, at least partially, Q. That is, to count as relevant, a sentence does not need to determine in which cell of Q the actual world is, but it must at least rule out one cell as a possible location.

The precise formulation of the above is captured in (1) below:

(1) ***Standard QUD Relevance***

Let S be a declarative sentence, $⟦S⟧$ its truth-conditions (the set of worlds in which S is true), C the context set (the set of worlds compatible with the common ground), Q a QUD (a partition of C), and q a cell of Q.

a. S is relevant w.r.t Q iff S gives a partial answer to Q;
b. S gives a partial answer to Q iff $⟦S⟧ \cap C \neq \emptyset$ and there is a $q \in Q$ s.t. $q \cap ⟦S⟧ = \emptyset$ (In English: S is not a contextual contradiction, and its truth-conditions fail to intersect, and hence they eliminate, at least one cell of Q.)[2]

I shall refer to (1)—which is nothing else but Groenendijk and Stokhof's (1984) definition of relevance[3] with some minor updates[4]—as *Standard QUD Relevance* (*SQR* for short).

Let's see how (1) works with an example:

(2) <u>A: Is Mary coming to the party tonight?</u>
B_1: She is (coming). [Relevant ✓]
B_2: Jeff Bezos hates Bill Gates. [Relevant ✗]

A's question induces a QUD, i.e. a two-cell partition—the YES-cell (the set of worlds in which Mary is coming to the party) and the NO-cell (the set of worlds in which she isn't coming). With respect to this QUD, (2)B_1 gives a partial answer and hence relevant: (2)B_1 is not a contextual contradiction (if it was common ground that Mary isn't coming, A wouldn't be asking such a question) and it eliminates the NO-cell, i.e. if (2)B_1 is the case, then it must be the case that the actual world lives somewhere in the

[2] The provision that a partial answer cannot be a (contextual) contradiction is present in all Groenendijk and Stokhof's definitions of partial answerhood I'm aware of—semantic ones, where question denotations are partitions of Logical Space, and pragmatic ones, where they are restricted to an information set (cf. Groenendijk and Stokhof 1984; 1990). If one disagrees with this provision, i.e. if one thinks that contradictions should count as relevant (cf. Bar-Lev 2024), then one can just drop '$⟦S⟧ \cap C \neq \emptyset$ and' from (1). The empirical issues that will be discussed here are completely orthogonal to this debate.
[3] See Groenendijk and Stokhof's (1984) Definition (38) on page 242, which is based on the notion of 'giving a partial pragmatic answer', whose actual definition appears in (28) on page 353. (Note: there is a typographical error in (28): '=' should be '$\subseteq$', as clarified in Groenendijk and Stokhof (1990: 22)—see Definition 21.2.)
[4] 'Question' talk is replaced by 'QUD' talk, QUDs are explicitly characterised as partitions of C, as in Roberts ([1996] 2012) and most subsequent literature, the notation is simplified, and the partial answerhood condition is expressed in a more concise manner.

YES-cell. (2)B_2, by contrast, doesn't give a partial answer and hence it is not relevant: (2)B_2 intersects both cells of the partition—i.e. it is compatible with Mary coming to the party, as well as with Mary not coming—and hence it eliminates the no cell.

SQR, albeit elegant, is rather limited—as observed, for example, in Simons et al. (2010: 316-7), it fails to diagnose relevance in cases in which the sentence merely raises the probability of the world being in some cell or other but doesn't rule out any given cell. Consider (3), for example.

(3) A: Is Mary coming to the party tonight?
B_1: She has an exam tomorrow morning. [Relevant ✓]
B_2: Jeff Bezos hates Bill Gates. [Relevant ✗]

It seems clear that, unlike (3)B_2, (3)B_1 is a relevant thing to say as a response to A's question. According to *SQR*, however, both (3)B_1 are (3)B_2 irrelevant: (3)B_1, just like (3)B_2, is compatible with both QUD cells and hence rules out none—indeed, it's possible for (3)B_1 to be true and for Mary not to attend the party; and it's possible for (3)B_1 to be true and for Mary to attend the party (she might not care about the exam, etc.). In other words, by uttering (3)B_1, the speaker communicates that Mary is unlikely to attend the party, but not that she will certainly not attend.[5]

To make sense of the contrast in (3), a probabilistic definition of relevance can be given, one that is, at some level of abstraction, conceptually analogous to *SQR*: *SQR*, abstracting away from technical details, entails that, if a sentence *S* is relevant w.r.t. a QUD, then *S* is not a contextual contradiction and is *informative* relative to the QUD, i.e. it does *something* to the QUD cells. In a non-probabilistic framework, 'being informative relative to the QUD' just means eliminating at least one QUD cell; in a probabilistic framework, 'being informative relative to the QUD' amounts, at minimum, to having a non-trivial effect on the probabilities of the QUD cells, i.e. to shifting the probabilistic weights among QUD cells. Thus, in a probabilistic framework, relevance can be defined in (4).

[5] This fact is exposed in full light in the following minimal pair:

(i) A: Is Mary coming to the party tonight?
B: She has an exam tomorrow morning. (✓) So, she probably won't.

(ii) A: Is Mary coming to the party tonight?
B: She is not. (#) So, she probably won't.

(4) ***Probabilistic QUD Relevance***

Let P a probability distribution over Ω (the set of all possible worlds[6]), Q a QUD (now defined as a partition of Ω[7]), c a cell of Q, S a sentence, and $⟬S⟭$ its truth-conditions.

S is relevant w.r.t to Q given P iff $P(⟬S⟭) \neq 0$, and there is an $c \in Q$ such that $P(c) \neq P(c \mid ⟬S⟭)$.
(In English: S is relevant w.r.t to Q given P iff S is not a contextual contradiction and learning that S is true shifts the probabilistic weights among *QUD* cells.)

To the best of my knowledge, the definition in (4), which I'll refer to as *PQR*, was first sketched in Büring (2003). It should be noted, however, that the idea of defining relevance in probabilistic terms is an old one. Carnap (1950), for example, characterised relevance as follows (with some minor notational modifications that preserve the original idea): a statement S is relevant w.r.t. a hypothesis H (a statement) and a distribution P iff $P(⟬S⟭) \neq 0$ and $P(⟬H⟭) \neq P(⟬H⟭|⟬S⟭)$. The only significant difference between this defintion and (4) is that, in Carnap's approach, the relevance of a statement is defined w.r.t. a statement, which expresses a proposition, and a probability distribution, whereas in (4) is defined w.r.t. a partition of a set of worlds (a set of propositions) and a distribution. The mechanics, however, are identical.

PQR, which is strictly weaker than *SQR*, predicts the contrast in (3), provided that the prior distribution is such that having an exam is a likely reason for Mary not to attend the party whereas Bezos hating Gates has no bearing on Mary's decision to attend. Indeed, given such a prior, (3)B_1 is bound to shift the probabilistic weights of the QUD cells (it is bound to raise the probability of the NO cell), whereas (3)B_2 will have no effect on the cell probabilities.

It is worth noting that, although *PQR* can account for the contrast in (3), it is not suited to explain, for example, why (3)B_1, though intuitively relevant, is not as relevant as, for example, (2)B_1. In other words, *PQR*, just like *SQR*, tells us whether a declarative sentence is relevant or not, but it doesn't tell us *how* relevant it is. To answer the latter question, a general method to assign a relevance score to a sentence/proposition is required. Such methods are readily available in the literature—see, for example, Russell (2012) for a scoring system based on Carnap's proposed measure of relevance, or Feinmann (2020) for a relevance scoring system based on Kullback–Leibler (KL) divergence. Here I won't be concerned with the issue of gradience in relevance judgments, so I won't discuss any of these methods.

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[6] For the ease of exposition, I'm assuming that Ω is finite.
[7] In Bayesian frameworks, the common ground, also known as 'the common prior', is identified with a probability distribution over Ω and not with a set of worlds. Thus, in this framework, QUDs are just partitions of Ω: those worlds that aren't compatible with the common ground, instead of being excluded from the QUD (as in the non-probabilistic framework), are simply assigned probability zero.

SQR and *PQR*, it should be stressed, can be conceptualised, at least at some level of abstraction, as different implementations of a single core notion, which I shall refer to as *Informativity-based QUD Relevance* (*Info-based QUD-R* for short). This notion can be summarised as follows:

(5) ***Informativity-based QUD Relevance***
  (i) The relevance of a sentence is determined relative to a QUD and an epistemic state.
  (ii) A sentence is relevant iff, given what is known/believed, it is *informative* w.r.t. the QUD, i.e. the proposition that it expresses has a non-trivial effect on the QUD cells.

In the *SQR* implementation, the epistemic state is standardly identified with the Stalnakerian context set (the set of worlds compatible with the common ground); in the *PQR*, in turn, the epistemic state is identified with the 'common prior' (a probability distribution).[8] In the *SQR* implementation, 'being informative' amounts to discarding cells, i.e. *a priori* possible ways of resolving the question; in the *PQR*-implementation, 'being informative' amounts to having a non-trivial effect on the probabilities of the QUD cells.

In the next section, I'll be presenting some data that call into question this conception of relevance.

2 Not all uninformative sentences are irrelevant

As mentioned, *Info-based QUD-R*, either if formalised as *SQR* or *PQR*, entails that the uttered sentence is doomed to irrelevance unless it is *informative* w.r.t. the QUD—unless it does something to the QUD cells. Now, let's assume that, according to the common prior, Mary is just as likely to come to the party as she is not to come—i.e. $P(\text{YES-cell}) = P(\text{NO-cell}) = 0.5$. Let's also assume, as we did before, that Bezos hating Gates has no bearing on Mary's decision to attend the party. Given this setup, consider the following contrast:

(6) <u>A: Is Mary coming to the party tonight?</u>
  B_1: Mary's father told me she's coming, but Mary's mother said the opposite. [Relevant ✓]
  B_2: Jeff Bezos hates Bill Gates. [Relevant ✗]

(6)B_1 seems like a relevant contribution given the question that has been raised: what B_1 is saying is clearly *related to* A's question, unlike the response in (6)B_2, which is clearly *unrelated*. This contrast, however, poses a serious challenge to the *Info-based QUD-R* approach. The issue is this: the first conjunct of (6)B_1 raises the probability of the QUD's YES-cell, whereas the second conjunct does precisely the opposite—it raises the probability of the QUD's NO-cell. Assuming there's no reason to

[8] See Morris (1995) for a discussion on the common prior assumption.

believe that one parent is more likely than the other to provide truthful information, learning that (6)B_1 is true is bound to have no net effect on the QUD cells: the conjuncts' respective effects on the probabilities simply cancel each other out. From *Info-based QUD-R*'s perspective, therefore, the contrast between (6)B_1 and (6)B_2 is unexpected: on this account, *both* utterances should be judged to be irrelevant.

The empirical point made by (6), it should be noted, has a historical precedent. Keynes (1921), to my knowledge, was the first to introduce the conception of relevance that lies at the heart of *PQR* and is often attributed to Carnap (1950), i.e. a statement S is relevant w.r.t. a hypothesis H (a statement) and a distribution P iff $P(\llbracket S \rrbracket) \neq 0$ and $P(\llbracket H \rrbracket) \neq P(\llbracket H \rrbracket | \llbracket S \rrbracket)$. Keynes, however, was unsatisfied with this notion; on his view, a statement of the form '$S_1 \wedge S_2$' should count as relevant to a given hypothesis H even if $P(\llbracket H \rrbracket) = P(\llbracket H \rrbracket | \llbracket S_1 \wedge S_2 \rrbracket)$ provided that $P(\llbracket H \rrbracket) \neq P(\llbracket H \rrbracket | \llbracket S_1 \rrbracket)$ and $P(\llbracket H \rrbracket) \neq P(\llbracket H \rrbracket | \llbracket S_2 \rrbracket)$.[9] This (conceptual) intuition, an intuition that is aligned with the judgments reported in (6), led Keynes (1921) to dismiss his own characterisation of relevance, which Carnap (1950) later revived.[10]

Now, if we look at (6)B_1 with a linguistic eye, the fact that (6)B_1 is intuitively relevant could be interpreted, not as fundamental problem for *Info-based QUD*-R as a relevance-assessing mechanism, but rather as an indication that this mechanism should be computed in stages, as interpretation proceeds, rather than on the global meaning of the uttered sentence. Indeed, what appears to be happening in (6)B_1 is this: first, one hears 'Mary's father told me she's coming', which, according to *PQR* (the probabilistic implementation of *Info-based QUD-R*), is relevant (it raises the probability of the QUD's YES-cell); then, one learns that 'Mary's mother told me she's coming', which, according to *PQR*, is also relevant (it raises the probability of the QUD's NO-cell). Thus, insofar as one uses *PQR*, and holds to the view that *PQR* can target different coordinate clauses as interpretation proceeds, one might not be too worried about (6)B_1—though a problem for *PQR* as it stands, it might be possible to provide a dynamic/incremental rendition of *PQR* that circumvents this problem.

The issue, however, appears to be deeper than what (6) suggests. Consider, for example, (7):

(7) <u>A: Is Mary coming to the party tonight?</u>
B_1: Mary's parent gave me conflicting reports. [Relevant ✓]
B_2: Jeff Bezos hates Bill Gates. [Relevant ✗]

[9] In Keynes' (1921: 72) own words: 'If we are able to treat "weight" and "relevance" as correlative terms, we must regard evidence as relevant, part of which is favourable and part unfavourable, even if, taken as a whole, it leaves the probability unchanged.'
[10] Anecdotally Keynes did formulate an alternative account of relevance, which Carnap proved untenable three decades later by showing it leads to trivialisation (see Carnap 1950: 419-20).

(7)B_1 has no effect on the QUD cells, nor does (7)B_2. Despite this, there's a clear contrast between (7)B_1 and (7)B_2: the former (just like (6)B_1) is intuitively relevant, while the latter is not. However, unlike (6)B_1, (7)B_1 is a simple sentence (it contains no sentential connectives). Since the problem posed by (7)B_1 is, as far as I can tell, the same problem that (7)B_1 poses, I don't see how a solution to this problem could come from formulating a dynamic/incremental version of *PQR*.[11]

3 *Info-based QUD-R*: final assessment

For Grice, if a linguistic contribution was related to 'the immediate needs of the transaction', then it was relevant to it, and vice versa—let's remember that he equates the Maxim of <u>Relation</u> with the command 'Be <u>relevant</u>'. In the early 1980s, Groenendijk and Stokhof came up with a very influential idea, that of conceptualising Grice's Maxim of Relation along the following lines: a linguistic contribution is relevant/related to the QUD (the topic of the conversation) iff it is informative to the QUD (in their framework, 'being informative' boils down to 'giving a partial answer'; see §1). Now, if one takes our observations at face value, it's clear that the notion of being relevant in the sense of being *related* cannot be assimilated to the notion of being relevant in the sense of being *informative*: a sentence may not be informative to the QUD (it may not have any detectable effect on the QUD cells) but may still be related to the QUD, as the contrast between (6)B_1 and (6)B_2 and that between (7)B_1 and (7)B_2 indicate.

This point can be amplified by considering the possible continuations that an uninformative-yet-relevant/related answer may allow *vis-à-vis* an uninformative-and-also-irrelevant/unrelated answer. Consider, for example, (8) and (9):

(8) <u>A: Is Mary coming to the party tonight?</u>
B: Mary's parents gave me conflicting reports. [uninformative & related to the QUD]

A_1: ✓ Well, that's not very helpful.
A_2: ?? And what does that have to do with anything?

(9) <u>A: Is Mary coming to the party tonight?</u>
B: Jeff Bezos hates Bill Gates. [uninformative & unrelated to the QUD]

A_1: ?? Well, that's not very helpful.
A_2: ✓ And what does that have to do with anything?

In the light of these data, I don't think that *Info-based QUD-R* can be said to successfully model Grice's Maxim of Relation; it fails to explain why (8)B is perceived as being related to the question raised

[11] Thanks to Daniel Rothschild for pressing on this point.

whereas (9)B is not. I would like to stress that I have no qualms with *Info-based QUD-R* as such—it is a perfectly natural notion, one that has empirical reflexes of its own, e.g. 'Well, that's not very helpful' works as a reply to the uninformative (8)B but wouldn't work as a reply to the informative (3)B_1 ('She has an exam tomorrow morning'). My point is simply that *Info-based QUD-R* rules out as irrelevant contributions that, albeit unhelpful or uninformative, do not give rise to a judgment of irrelevance.

As I see it, the data discussed here can be interpreted in at least two ways:

(i) *Info-based QUD-R* is not the right approach to modelling linguistic relevance.

(ii) *Info-based QUD-R* is an important ingredient of 'the relevance algorithm' but is not the only ingredient. On this view, *Info-based QUD-R* does play a role in the production of our judgments of relevance *in conjunction with* other considerations, which are yet to be elucidated.

At this point, I don't have any reason to favour one interpretation over another. That said, in these final pages, I would like to consider a rather natural conjecture that falls under (ii)—namely, that our judgments of relevance, though dictated primarily by *Info-based QUD-R,* are also modulated by what one may call 'QUD accommodation', i.e. the mechanism whereby the QUD induced by the overt question is 'replaced' by another QUD.

3.1 Invoking other questions?

Indeed, one possible theory could have *Info-based QUD-R* as a central component, alongside an account of QUD accommodation. According such a theory, (8)B, repeated below as (10)B, would be perceived as relevant because it would not be interpreted as an answer to the QUD induced by the overt question but as an answer to an accommodated QUD, one relative to which *Info-based QUD-R* is satisfied. Conversely, (9)B, repeated below as (11)B, would be perceived as irrelevant because it would not be possible to accommodate a QUD relative to which *Info-based QUD-R* is satisfied.

(10) A: Is Mary coming to the party tonight?
B: Mary's parents gave me conflicting reports. [uninformative & related to the QUD]

(11) A: Is Mary coming to the party tonight?
B: Jeff Bezos hates Bill Gates. [uninformative & unrelated to the QUD]

It is not clear to me how to spell out a constraint on QUD accommodation that accomplishes this. (10)B could be taken to address the QUD in (12): (10)B is clearly informative relative to such a QUD—it eliminates those cells in which Mary's parents agree on their guesses—so *Info-based QUD-R* (either *SQR* or *PRQ*) is satisfied.

(12) What are Mary's parents' best guesses as to whether Mary is attending the party?

{{mother and father think she's coming}, {mother and father think she's not coming}, {mother thinks she's coming, and father thinks she's not}, {father thinks she is coming, and mother thinks she is not}, etc.[12]}

However, by the same token, (11)B could be taken to address the QUD in (13), which would lead to a bad prediction: it would predict (11)B to be perceived as relevant, contrary to fact.

(13) Does Jeff Bezos hate Bill Gates?

{{jeff bezos hates bill gates}, {jeff bezos doesn't hate bill gates}}

To address this problem, one could put forth the following constraint on QUD accommodation (in the following definition, and in the paragraphs that follow, I'll be working with *PQR*, the probabilistic implementation of *Info-based QUD-R*[13]):

(14) Let S be the uttered sentence, Q the QUD induced by the overt question, and Q^* any other QUD. If learning that S is the case doesn't shift the probabilistic weights among Q-cells—that is, if S is diagnosed as irrelevant by *PQR*—it is possible to dismiss Q and accommodate Q^* provided that:

(i) Learning that S is the case shifts the probabilistic weights among Q^*-cells.

(ii) There is a cell $c \in Q^*$ such that c shifts the probabilistic weights among Q-cells.

The constraint in (14) does block the QUD in (13) from being accommodated in (11). Quite clearly, there's no cell in (13) that shifts the probabilistic weights among the cells of the QUD induced by (11)A—whether Jeff Bezos hates Bill Gates, or whether he doesn't, is orthogonal to the question of whether Mary is coming to the party of not.

(14), by contrast, doesn't interfere with the possibility of accommodating (12) in (10). Indeed, in this case, there's a cell in (12) (e.g. {mother and father think she's not coming}) that shifts the probabilistic weights among the cells of the QUD induced by (10)A.

[12] To avoid clutter, I'm not listing cells in which at least one of the parents is maximally uncertain about Mary's plans.

[13] If one opts to work with *SQR* (the non-probabilistic implementation of *Info-based QUD-R*), one has not one but two problems—i.e. in addition to explaining the contrast between (8) and (9), one also needs to account for the contrast in (3).

The issue, however, is this: there are in principle many QUDs that could be accommodated in (11) to force (11)B into *PQR* satisfaction. The constraint in (14) does prevent the QUD in (13) from being accommodated but, for example, is unable to filter out a QUD such as the one in (15).

(15) {{jeff bezos hates bill gates and mary has an exam tomorrow morning}, {jeff bezos hates bill gates and mary doesn't have an exam tomorrow morning}, {jeff bezos doesn't hate bill gates and mary has an exam tomorrow morning}, {jeff bezos doesn't hate bill gates and mary doesn't have an exam tomorrow morning}}

Indeed, learning that (11)B is true shifts the probabilistic weights among the cells of (15)—so condition (14)i is met; furthermore, there is a cell *c* in (15) (e.g. {jeff bezos hates bill gates and mary has an exam tomorrow morning}) that shifts the probabilistic weights of the QUD cells induced by the overt question (*Is Mary coming to the party?*)—so condition (14)ii is also met.

(14) is therefore too weak. A stronger constraint on QUD accommodation would be needed—one that prevents QUD accommodation from rescuing (11)B while allowing this mechanism to rescue (10)B. Whether such a constraint can be formulated, I don't know—this is something I must leave for future research to determine.

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